

● HARD

● GEOMETRY AND TRIGONOMETRY

Right triangles and trigonometry

01

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

$$RS = 440$$

$$ST = 384$$

$$TR = 584$$

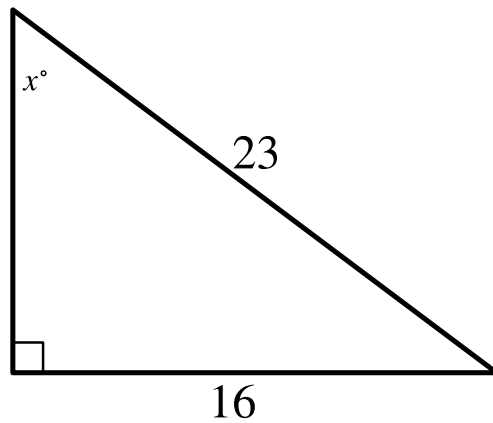
The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

A. $\frac{48}{73}$

B. $\frac{55}{73}$

C. $\frac{48}{55}$

D. $\frac{55}{48}$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The length of the hypotenuse is 23.
- The length of the leg opposite the angle with measure x° is 16.
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x^\circ$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

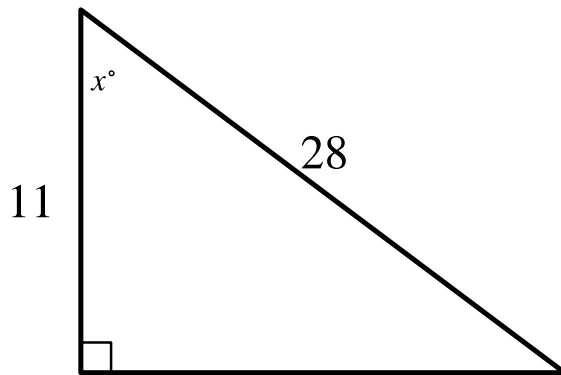
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

For two acute angles, $\angle Q$ and $\angle R$, $\cos Q = \sin R$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$, respectively. What is the value of x ?

- A. 5
- B. 19
- C. 23
- D. 29

Which of the following expressions is equivalent to $\sin 24^\circ \cos 66^\circ + \cos 24^\circ \sin 66^\circ$?

- A. $2\cos 66^\circ \sin 24^\circ$
- B. $2\cos 66^\circ + 2\cos 24^\circ$
- C. $\cos 66^{\circ 2} + \cos 24^{\circ 2}$
- D. $\cos 66^{\circ 2} + \sin 24^{\circ 2}$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The length of the hypotenuse is 28.
- The length of the leg adjacent to the angle with measure x° is 11.
- A note indicates the figure is not drawn to scale.

In the triangle shown, what is the value of $\cos x^\circ$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

GRID-IN

GEOMETRY AND TRIGONOMETRY

A triangle with angle measures 30° , 60° , and 90° has a perimeter of $18+6\sqrt{3}$. What is the length of the longest side of the triangle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angle C corresponds to angle F . Angles C and F are right angles. If $\tan A = \frac{50}{7}$, what is the value of $\tan E$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN

GEOMETRY AND TRIGONOMETRY

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1,089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?

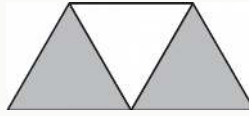
STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In triangle ABC , angle B is a right angle. The length of side AB is $10\sqrt{37}$ and the length of side BC is $24\sqrt{37}$. What is the length of side AC ?

- A. $14\sqrt{37}$
- B. $26\sqrt{37}$
- C. $34\sqrt{37}$
- D. $\sqrt{34 \cdot 37}$



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

- A. $2\sqrt{3}$
- B. $4\sqrt{3}$
- C. $8\sqrt{3}$
- D. 16